

Model Reduction by Moment Matching for Linear Singular Systems

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Abstract—The paper presents a moment matching approach to the model reduction problem for singular systems. Combining the interpolation-based and the steady-state-based description of moment, a partitioned formulation of the Krylov projector is obtained. Several implications of this result are investigated and different families of reduced order models are proposed. The possibility to maintain structural properties of system is studied. Two examples illustrate the results of the paper.

I. INTRODUCTION

Singular systems, also known as generalized state space systems, descriptor systems or differential-algebraic equations, have been largely investigated because of their capacity of modeling a large variety of physical behaviors such as power systems, electrical networks, chemical processes, biological systems and social economic systems, see [1], [2]. The important characteristic of this class of systems is that they combine dynamic (differential) equations, referred as *slow subsystem* and static (algebraic) equations, referred as *fast subsystem*. They are often the result of the interconnection of several subsystems and the overall singular system has often large dimensionality. This justifies the interest in the model reduction problem for this class of systems. A first result of model reduction for singular systems has been proposed in [3] in which the chained aggregation method has been used. As pointed out in [4] this method is computationally intensive and another approach based on balanced realization has been proposed. However, the reduced order model obtained therein is a non-singular one and the technique is not able to maintain the impulsive characteristic of singular systems. In [5] a singular reduced order model based on the Nehari's approximation algorithm [6] has been presented. Other results of model reduction via covariance approximation [7] and using singular value decomposition [8] have also been proposed. In [9] the authors propose a moment matching method to reduce multi-input, multi-output (MIMO) systems. The method proposed therein is based in isolating the slow and fast subsystem and in reducing only the slow part (the non-singular subsystem), leaving unreduced the fast part (the singular subsystem). In this paper we follow the moment matching approach, see *e.g.* [10], [11], [12], [13], [14], [15] and [16]. In particular we extend the model reduction method developed in [17] to singular systems. This method has the ability to preserve in a direct way some properties of the original system, overcoming some of the drawbacks of the moment matching

methods. Moreover, it can be extended to nonlinear systems, see *e.g.* [18], [19] and [20]. Combining the interpolation-based and the steady-state-based description of moment (see [21] and [22] for a detailed discussion of the two descriptions) a partitioned projector Π is computed. In the projector, the contribution of the fast subsystem is separated by the contribution of the slow subsystem. The information on the fast subsystem is clearly encoded in the projector and it is not lost by the moment matching technique. Moreover, the output of reduced order models based on this projector approximates the output of the system only when consistent initial conditions are taken in account. In addition, the partition of the projector allows to define several families of reduced order models which have great flexibility in maintaining independently specific properties of the slow and fast subsystems. Thus, purely fast, purely slow and a “simple” family of reduced order models are proposed and the possibility of matching with impulsive controllability constraints is discussed.

Remark 1: The contribution of this paper is substantially different from [9], in both results and meaning. In [9] the fast subsystem is not reduced at all. The reason the authors give is that if the algebraic part is reduced than the frequency error between the system to be reduced and the reduced order model grows unbounded. Thus, their projector is partitioned in a classical projector (for non-singular systems) and the identity. In the method proposed here the reduction can operate independently on the algebraic part, on the differential part, or on a combination thereof. Moreover, we believe that the ability of reducing the algebraic part is relevant even though the error will grow unbounded. In fact, as shown in the example in Section V-A, a reduced order model which has the same output response of the system to be reduced can be determined even though for high frequencies the error grows unbounded. Moreover, in many applications the unbounded part is filtered out with a low-pass filter or, alternatively, the behavior for high frequencies is not of interest as long as the instability is maintained (both the system and the reduced order model grow unbounded). Finally note that the results in this paper are instrumental in deriving a model reduction theory for nonlinear singular systems.

The rest of the paper is organized as follows. In Section II we recall some basic results for singular systems which are instrumental to develop the remaining of the paper. In Section III the interpolation-based description of moment is extended to singular systems. Then, combining it with the steady-state-based description of moment, a new formulation

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of the projector Π is given. In Section IV several families of reduced order models are given and the possibility to retain, reduce or eliminate the fast subsystem in the reduced order model is investigated. Moreover, a result to enforce impulsive controllability on the reduced order model is proposed. In Section V two numerical examples are used to show the application of the results of the paper. Finally Section VI contains some concluding remarks and future directions of research.

Notation. We use standard notation. $\mathbb{C}_{<0}$ denotes the set of complex numbers with negative real part, whereas $\emptyset$ indicates the empty set. The symbol I denotes the identity matrix and $\sigma(A)$ denotes the spectrum of the matrix $A \in \mathbb{R}^{n \times n}$. The superscript $\top$ denotes the transpose.

II. PRELIMINARIES ON SINGULAR SYSTEMS

In this section some basic results for singular systems are recalled. Consider a linear, single-input, single-output, continuous-time, singular system described by the equations¹

$$E\dot{x} = Ax + Bu, \quad y = Cx, \quad (1)$$

with $x(t) \in \mathbb{R}^n$, $u(t) \in \mathbb{R}$, $y(t) \in \mathbb{R}$, $E \in \mathbb{R}^{n \times n}$, $A \in \mathbb{R}^{n \times n}$, $B \in \mathbb{R}^{n \times 1}$ and $C \in \mathbb{R}^{1 \times n}$. Assume that $u(t)$ is piecewise continuously differentiable.

Definition 1: [1] Let $\mathcal{E} \subset \mathbb{C}$ be the set² of complex numbers s such that $\det(sE - A) \neq 0$. For any two matrices E and A , the pencil (E, A) is called *regular* if $\mathcal{E} \neq \emptyset$.

Lemma 1: [1] The pencil (E, A) is regular if and only if there exist two nonsingular matrices Q and P such that

$$QEP = \text{diag}(I, N), \quad QAP = \text{diag}(A_1, I),$$

where $N \in \mathbb{R}^{n_2 \times n_2}$ is nilpotent with degree³ h and $A_1 \in \mathbb{R}^{n_1 \times n_1}$ is full rank, with $n_1 + n_2 = n$.

Assume system (1) is regular, then Q and P can be selected such that system (1) can be written in the so-called *first equivalent form*, namely

$$\begin{array}{ll} \text{slow subsystem:} & \text{fast subsystem:} \\ \dot{x}_1 = A_1 x_1 + B_1 u, & N \dot{x}_2 = x_2 + B_2 u, \\ y_1 = C_1 x_1, & y_2 = C_2 x_2, \\ y = y_1 + y_2, & \end{array} \quad (2)$$

with the coordinate transformation $\begin{bmatrix} x_1^\top & x_2^\top \end{bmatrix}^\top = P^{-1}x$, where $x_1 \in \mathbb{R}^{n_1 \times n_1}$, $x_2 \in \mathbb{R}^{n_2 \times n_2}$, and

$$\begin{array}{ll} QEP = \text{diag}(I, N), & QB = \begin{bmatrix} B_1^\top & B_2^\top \end{bmatrix}^\top, \\ QAP = \text{diag}(A_1, I), & CP = \begin{bmatrix} C_1 & C_2 \end{bmatrix}. \end{array}$$

¹The results can be extended to MIMO systems straightforwardly. The problem in the MIMO case is called tangential interpolation and is discussed, in the present context, in [23], see also [24].

²See [1] for a characterization of this set.

³A nilpotent matrix is a square matrix N such that $N^k = 0$ for some positive integer k . The smallest such k is called the degree of N .

The state response of system (1) is given by

$$\begin{aligned} x(t) = P \begin{bmatrix} I \\ 0 \end{bmatrix} & \left(e^{A_1 t} x_1(0) + \int_0^t e^{A_1(t-\tau)} B_1 u(\tau) d\tau \right) \\ & - P \begin{bmatrix} 0 \\ I \end{bmatrix} \sum_{i=0}^{h-1} N^i B_2 \frac{d^i}{dt^i} u(t). \end{aligned} \quad (3)$$

Differently from standard linear systems, singular systems have a unique solution only if $x(0) = P \begin{bmatrix} x_1^\top(0) & x_2^\top(0) \end{bmatrix}^\top$ is a *consistent* initial condition, namely it is such that

$$x(0) = P \begin{bmatrix} I \\ 0 \end{bmatrix} x_1(0) - P \begin{bmatrix} 0 \\ I \end{bmatrix} \sum_{i=0}^{h-1} N^i B_2 \frac{d^i}{dt^i} u(t) \Big|_0,$$

where $x_1(0)$ can be freely selected.

III. DEFINITION OF MOMENT

In this section we establish a one-to-one relation between the moments of system (1), the unique solution of a generalized Sylvester equation and the steady-state response of the output of a particular interconnected system.

Let

$$\begin{aligned} W(s) &= C(sE - A)^{-1}B \\ &= C_1(sI - A_1)^{-1}B_2 + C_2(sN - I)^{-1}B_2, \end{aligned}$$

be the transfer function associated to system (1) and assume that the system is minimal, i.e., as for non-singular systems (see [1, Theorem 2-6.3]), controllable and observable.

A. Interpolation-based description of moment

We begin extending the classical interpolation-based description of moment. This, in conjunction with the steady-state-based description of moment, allows to obtain a special structure for the projector to be used in the model reduction process.

Definition 2: Let $s_i \in \mathcal{E}$. The *0-moment of system (1) at s_i* is the complex number $\eta_0(s_i) = C(s_i E - A)^{-1}B$. The *k-moment of system (1) at s_i* is the complex number

$$\eta_k(s_i) = \frac{(-1)^k}{k!} \left[\frac{d^k}{ds^k} W(s) \right]_{s=s_i},$$

with $k \geq 1$ integer.

Note that each moment is the sum of two contributions, one depending on the slow subsystem and one depending on the fast subsystem, i.e.

$$\begin{aligned} \eta_k(s_i) &= (-1)^k C_1(s_i I - A_1)^{-(k+1)} B_1 \\ &\quad + (-1)^k C_2 N^k (s_i N - I)^{-(k+1)} B_2, \end{aligned} \quad (4)$$

and if $k \geq h$,

$$\eta_k(s_i) = (-1)^k C_1(s_i I - A_1)^{-(k+1)} B_1.$$

This definition of moment is justified by the fact that the *k-moment of a linear system at s_i* is defined as the *k-th coefficient of the Laurent series expansion of the transfer*

function $W(s)$ at $s_i \in \mathbb{C}$ (see [10, Chapter 11]), provided it exists. The next result, which is a direct extension of [17] (see also [25] and [15]), gives a relation between the moments and the solution of a generalized Sylvester equation.

Lemma 2: Let $s_i \in \mathcal{E}$. Consider system (1), then

$$\begin{bmatrix} \eta_0(s_i) & \dots & \eta_k(s_i) \end{bmatrix} = C\tilde{\Pi}\Psi_k,$$

where $\Psi_k = \text{diag}(1, -1, 1, \dots, (-1)^k) \in \mathbb{R}^{(k+1) \times (k+1)}$ and $\tilde{\Pi}$ is the unique solution of the generalized Sylvester equation

$$A\tilde{\Pi} + BL_k = E\tilde{\Pi}\Sigma_k, \quad (5)$$

with $L_k = [1 \ 0 \ \dots \ 0] \in \mathbb{R}^{(k+1)}$ and

$$\Sigma_k = \begin{bmatrix} s_i & 1 & 0 & \dots & 0 \\ 0 & s_i & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & \dots & 0 & s_i & 1 \\ 0 & \dots & \dots & 0 & s_i \end{bmatrix} \in \mathbb{R}^{(k+1) \times (k+1)}.$$

Exploiting this lemma, the following result holds.

Theorem 1: Consider system (1) and suppose that for a set of numbers s_i , with $i = 1, \dots, \eta$, $s_i \in \mathcal{E}$. Then there exists a one-to-one relation between the moments $\eta_0(s_1), \dots, \eta_{k_1-1}(s_1), \dots, \eta_0(s_\eta), \dots, \eta_{k_\eta-1}(s_\eta)$ and the matrix $C\Pi$, where Π is the unique solution of the generalized Sylvester equation

$$A\Pi + BL = E\Pi S, \quad (6)$$

with $S \in \mathbb{R}^{\nu \times \nu}$ any non-derogatory matrix with characteristic polynomial

$$p(s) = \prod_{i=1}^{\eta} (s - s_i)^{k_i}, \quad (7)$$

where $\nu = \sum_{i=1}^{\eta} k_i$ and L such that the pair (L, S) is observable.

B. Steady-state-based description of moment

In this section we present the steady-state-based description of moment for linear singular systems. This is instrumental to derive the main technical result of the paper, namely a partition of the projector Π in slow and fast parts.

Theorem 2: Let $S \in \mathbb{R}^{\nu \times \nu}$ be any non-derogatory matrix with characteristic polynomial (7). Consider system (1) and suppose that $\sigma(S) \subset \mathcal{E}$ and $\sigma(A) \subset \mathbb{C}_{<0}$. Consider the interconnection of system (1) with the system

$$\dot{\omega} = S\omega, \quad u = L\omega, \quad (8)$$

with L and $\omega(0)$ such that the triple $(L, S, \omega(0))$ is minimal. Then, for all consistent initial conditions, there exists a one-to-one⁴ relation between the moments $\eta_0(s_1), \dots, \eta_{k_1-1}(s_1), \dots, \eta_0(s_\eta), \dots, \eta_{k_\eta-1}(s_\eta)$ and the steady-state response of the output of such interconnected system.

⁴By one-to-one relation we mean that the moments are uniquely determined by the steady-state response of $y(t)$ and vice versa.

We are ready now to give the main result of the paper.

Theorem 3: Let $S \in \mathbb{R}^{\nu \times \nu}$ be any non-derogatory matrix with characteristic polynomial (7). Consider system (2) and suppose that $\sigma(S) \cap \sigma(A_1) = \emptyset$. Then there exists a one-to-one relation between the moments $\eta_0(s_1), \dots, \eta_{k_1-1}(s_1), \dots, \eta_0(s_\eta), \dots, \eta_{k_\eta-1}(s_\eta)$ and the matrix $C\hat{\Pi}$, where $\hat{\Pi} = P^{-1}\Pi = [\Pi_1^\top \ \Pi_2^\top]^\top$ is the unique solution of the equations

$$\begin{aligned} A_1\Pi_1 - \Pi_1 S &= -B_1 L, \\ \Pi_2 &= -\sum_{i=0}^{h-1} N^i B_2 L S^i. \end{aligned} \quad (9)$$

Remark 2: The partition given in Theorem 3 can be computed directly using the interpolation-based description of moment given in Theorem 1. However, the proof of the result is easier if the steady-state-based description of moment is exploited.

This result has multiple implications. The contribution to the moment of the fast subsystem is separated in the projector $\hat{\Pi}$ from the contribution of the slow subsystem. The condition for uniqueness of the projector is simplified and it depends only on the eigenvalues of the slow subsystem. This is actually expected since in the first equivalent form the k -th moment of the system can be written as (4), in which $s_i N - I$ is full-rank for any s_i .

It is clear from the form of $\hat{\Pi}$ that the information on the fast subsystem is encoded in the projector and it is not lost by the moment matching technique. Thus, the output of reduced order models based on this projector approximates the output of the system only when consistent initial conditions are taken in account, i.e. when the solution of the system to be reduced exists. Moreover, this formulation gives high flexibility in maintaining the properties of the fast and slow subsystems. Depending on the particular application the fast subsystem can be preserved, reduced or eliminated independently of the reduction of the slow subsystem. This possibility is analyzed in detail in the next section.

IV. REDUCED ORDER MODELS

In this section several families of reduced order models achieving moment matching are presented. The possibility of obtaining purely fast or purely slow singular systems is investigated and a “simple” family of singular systems is given. In addition, the solution to the problem of matching with impulsive controllability is given.

Definition 3: Consider system (1) and let $S \in \mathbb{R}^{\nu \times \nu}$ be any non-derogatory matrix with characteristic polynomial (7). Assume $\sigma(S) \cap \sigma(A_1) = \emptyset$ and let L be such that the pair (L, S) is observable. Then the system

$$\begin{aligned} \dot{\xi}_1 &= F_1 \xi_1 + G_1 u, & M \dot{\xi}_2 &= \xi_2 + G_2 u, \\ \psi_1 &= H_1 \xi_1, & \psi_2 &= H_2 \xi_2, \\ \psi &= \psi_1 + \psi_2, \end{aligned} \quad (10)$$

with $\xi_j(t) \in \mathbb{R}^\nu$, for $j = 1, 2$, $\psi_j(t) \in \mathbb{R}$, for $j = 1, 2$, $F_1 \in \mathbb{R}^{\nu \times \nu}$ full-rank, $M \in \mathbb{R}^{\nu \times \nu}$ nilpotent with degree h ,

$G_j \in \mathbb{R}^{\nu \times 1}$, for $j = 1, 2$, and $H_j \in \mathbb{R}^{1 \times \nu}$, for $j = 1, 2$, is a model of system (1) at S , if there exists a unique solution $\begin{bmatrix} \Gamma_1^\top & \Gamma_2^\top \end{bmatrix}^\top$ of the equations

$$\begin{aligned} F_1 \Gamma_1 - \Gamma_1 S &= -G_1 L, \\ \Gamma_2 &= -\sum_{i=0}^{h-1} M^i G_2 L S^i, \end{aligned} \quad (11)$$

such that

$$\begin{aligned} C_1 \Pi_1 &= H_1 \Gamma_1, \\ C_2 \Pi_2 &= H_2 \Gamma_2, \end{aligned} \quad (12)$$

where $\begin{bmatrix} \Pi_1^\top & \Pi_2^\top \end{bmatrix}^\top$ is the unique solution of (9). System (10) is a **reduced order model of system (1) at S if $2\nu < n$** .

We analyze now some special families of reduced order models which achieve moment matching.

1) Non-singular reduced order model: using the results of the paper, it is easy to approximate a singular system with a non-singular system. This was the first approach attempted, see [4]. The same authors and following researchers, e.g. see [5], deemed the method inaccurate, since the impulsive characteristic of the singular system is lost. However, in particular settings and as a first approximation it can be useful to approximate a singular system with a non-singular one. This can be done easily in the presented framework. In fact, the system

$$\dot{\xi} = F\xi + Gu, \quad \psi = H\xi, \quad (13)$$

is a model of system (1) at S , if there exists a unique solution Γ of the equation

$$F\Gamma - \Gamma S = -GL, \quad (14)$$

such that

$$C\Pi = H\Gamma, \quad (15)$$

where Π is the unique solution of (6). Note that, the reduced order model (13) has dimension ν , whereas the reduced order model (10) has dimension 2ν .

2) Fast reduced order model: in the same way, a completely fast reduced order model which approximates the singular system can be obtained in the presented framework. The system

$$M\dot{\xi} = \xi + Gu, \quad \psi = H\xi, \quad (16)$$

is a model of system (1) at S , if

$$H = -C\Pi \left(\sum_{i=0}^{h-1} M^i G L S^i \right)^{-1}, \quad (17)$$

where Π is the unique solution of (6), is uniquely determined. The reduced order model (16) has dimension ν .

3) The identity family of singular reduced order models: as already shown in [17], see also [19], a simple family of reduced order models is identified by the choice $\Gamma = I$. In the present setting it seems natural to select $\Gamma_1 = I$ keeping M and G_2 free. Then, the system

$$\begin{aligned} \dot{\xi}_1 &= (S - G_1 L)\xi_1 + G_1 u, \\ M\dot{\xi}_2 &= \xi_2 + G_2 u, \\ \psi_1 &= C_1 \Pi_1 \xi_1, \\ \psi_2 &= C_2 \sum_{i=0}^{h-1} N^i B_2 L S^i \left(\sum_{i=0}^{h-1} M^i G_2 L S^i \right)^{-1} \xi_2, \\ \psi &= \psi_1 + \psi_2, \end{aligned} \quad (18)$$

where $\begin{bmatrix} \Pi_1^\top & \Pi_2^\top \end{bmatrix}^\top$ is the unique solution of (9), is a model of system (1) at S for any G_1 , G_2 and M such that (11) has a unique solution.

Remark 3: The family of systems (18) belongs to the family given in Definition 3. The families (13) and (16) do not belong to the family given in Definition 3.

4) Matching with impulsive controllability: impulse controllability is a characteristic property of singular systems and corresponds to the controllability of only the fast subsystem. A condition for impulse controllability is given in [1]. System (1) is impulse controllable if and only if

$$\text{rank} \begin{bmatrix} E & 0 & 0 \\ A & E & B \end{bmatrix} = n + \text{rank}(E). \quad (19)$$

If the system to be reduced is impulsive controllable, it is desirable to maintain this property also in the reduced order model.

Theorem 4: System (10), or (16), or (18), is impulsive controllable if M and G_2 are selected such that

$$\text{rank} \begin{bmatrix} M & G_2 \end{bmatrix} = \text{rank}(M). \quad (20)$$

The proof of the theorem is straightforward, however, to the best of the author's knowledge, the condition for impulsive controllability has not been presented in the simple form given in (20).

Remark 4: As noted in [9], in the reduction of non-proper $W(s)$, namely systems with a nilpotent degree larger than zero, the algebraic part should be maintained unchanged to guarantee that the error between the system to be reduced and the reduced order model stays bounded. This is obviously implied by the fact that the absolute value of the frequency response grows unbounded as the frequency increases and the slope of this growth depends on the degree of the algebraic part of the transfer function. Thus, to have a small H_2 error norm the degree of the algebraic part should be maintained unaltered. However, in many cases we are not interested in preserving the rate at which the frequency response grows unbounded, but we are interested only in the maintaining the property that it grows unbounded. Or alternatively, we may be interested only in the output response for low frequencies. In this cases, the method in [9]

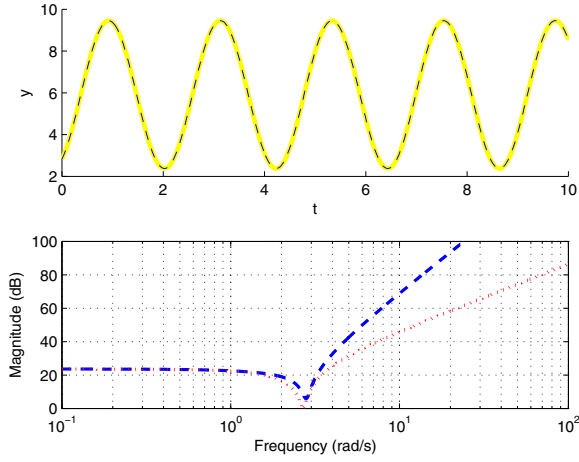


Fig. 1. (Top) Time history of the output of system (21) (solid line) and of the output of the reduced order model (22) (dashed line). (Bottom) Magnitude of the Bode plot of system (21) (dashed line) and of the reduced order model (22) (dotted line).

is not able to reduce the fast subsystem. This limit is clearly averted with this approach which leaves to the designer the choice of reducing the fast subsystem, the slow subsystem, or combination thereof.

Remark 5: The interpolation points, namely the eigenvalues of S , should be chosen depending on the specific application, analysis or control objective. However, an optimal choice with respect to the H_2 norm is described in [9].

V. EXAMPLES

In this section we use two numerical examples to illustrate the results of the paper. The first example is a purely fast system used often in the literature of model reduction of singular systems [4], [5]. The second example is a large-scale randomly generated singular system.

A. A classical fast subsystem example

This example has been proposed in [4], see also [5]. Consider the purely fast fifth order singular system described by the equations

$$N\dot{x} = x + Bu, \quad y = Cx, \quad (21)$$

with

$$N = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0.1 \\ 0.2 \\ 1.8 \\ 2.5 \\ 3.0 \end{bmatrix}, \quad C^T = \begin{bmatrix} 0.1 \\ 0.3 \\ 1.2 \\ 1.8 \\ 2.8 \end{bmatrix}.$$

Let $\sigma(S) = \{0, \pm 2.85j\}$ and L and $\omega(0)$ in (8) be randomly generated. Then, system (16) is a third order reduced order model of system (21), namely

$$M\dot{\xi} = \xi + Gu, \quad \psi = C \sum_{i=0}^{h-1} N^i B L S^i \left(\sum_{i=0}^{h-1} M^i G L S^i \xi \right)^{-1} \xi, \quad (22)$$

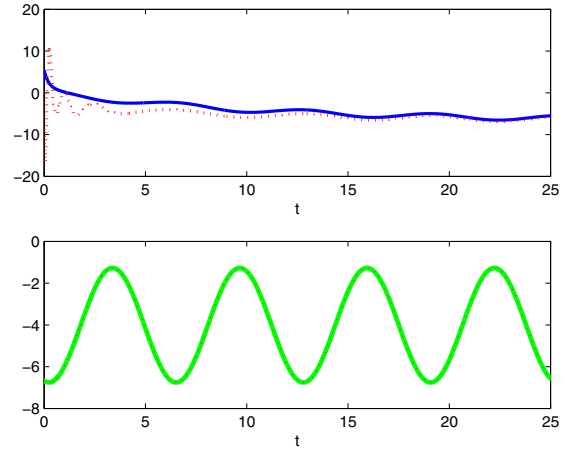


Fig. 2. (Top) Time history of the output of the slow subsystem (solid line) and the output of the slow subsystem of the reduced order model (dotted line). (Bottom) Time history of the output of the fast subsystem (solid line) and the output of the fast subsystem of the reduced order model (dotted line).

where G is randomly generated and

$$M = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

Fig. 1 (top) shows the time history of the output of system (21) (solid line) and of the output of the reduced order model (22) (dashed line). The two responses overlap because a purely fast system does not have a transient response. Fig. 1 (bottom) shows the magnitude of the Bode plot of system (21) (dashed line) and of the reduced order model (22) (dotted line). This plot gives a more indicative information of the quality of the approximation. In fact, from the figure we see that the reduced order model is a good approximation of the system at low frequencies ($< \approx 3$ rad/s).

Remark 6: As anticipated the method proposed in [9] cannot be used to reduce system (21). With the method proposed herein a reduced order model is computed and the output response of system (21) is perfectly matched. Note also that increasing the number of interpolation points would reduce the discrepancy shown in the bottom graph of Fig. 1. When the dimension of the reduced order model reaches the dimension of the system, the error will be zero.

B. A large-scale singular system

Consider the singular system (2) with $n_1 = 60$ and $n_2 = 20$. Let $A_1 = \text{diag}(-0.1, -0.2, \dots, -6.0)$, N be a canonical nilpotent matrix of degree $h = 20$ and B_1, B_2, C_1, C_2 and L be randomly generated. The matrix S is such that $\sigma(S) = \{0, \pm 0.1j, \pm 0.5j, \pm j, \pm 1.13j, \pm 5j\}$. The reduced order model (18) is computed. The matrix G_1 is selected such that $\sigma(S - G_1 L) \subset \mathbb{C}_{<0}$, M is a canonical nilpotent matrix of degree $h = 10$ and G_2 is selected as a truncation of B_2 . Fig. 2 (top) shows the time history of the output of the slow subsystem (solid line) and the output of the slow

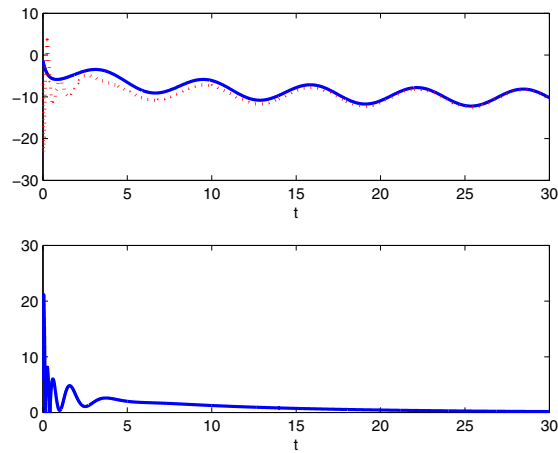


Fig. 3. (Top) Time history of the total output of the system (solid line) and of the reduced order model (dotted line). (Bottom) Absolute error between the two output responses.

subsystem of the reduced order model (dotted line). Fig. 2 (bottom) shows the time history of the output of the fast subsystem (solid line) and the output of the fast subsystem of the reduced order model (dotted line). As expected the bottom plots overlap. Fig. 3 (top) shows the time history of the total output y of the system (solid line) and of the total output ψ of the reduced order model (dotted line). Fig. 3 (bottom) shows the absolute error between the two output responses. As expected the error between the two responses depends only on the response of the slow subsystem and goes to zero as the transient vanishes.

VI. CONCLUSION

The model reduction technique based on moment matching has been extended to singular systems. Combining the interpolation-based and the steady-state-based description of moment a partitioned projector is constructed. The contribution of the slow subsystem and the contribution of the fast subsystem to the moment are separated. It is clear that the information on the fast subsystem is encoded in the projector and it is not lost by the moment matching technique. Moreover, the output of reduced order models based on this projector approximates the output of the system only when consistent initial conditions are taken in account. Exploiting this partitioned projector, several families of reduced order models have been proposed. In particular, a purely fast, a purely slow and a “simple” family of reduced order models have been proposed. The possibility of maintaining the impulsive controllability property has been investigated and a few examples have been used to illustrate the results. A natural extension of the current results is represented by the model reduction of nonlinear singular system and by the model reduction of systems with constraints.

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